

Climate 5000 Series - Gen 4

115V - 12K BTU Single Zone Regular System

Wall Mounted IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

115V - 12K Btu Single Zone Regular System - Wall Mounted IDU

Indoor part number		8-733-962-670
Indoor model number		BMS500-AAS012-0AHWXD
Outdoor part number		8-733-962-671
Outdoor model number		BMS500-AAS012-0CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		23.4
EER2		13
HSPF2 Region IV		9.5
HSPF2 Region V		7.0
COP(47F)		3.70
COP (5F)		2.33
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	2600~13000
	Rated Capacity	12000
	Current	10.3
	Input Power	923
Heating	Capacity Range	4600~13600
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	7700
	Current	10.5
	Input Power	950
Electrical Data		
Power supply (V / Ø / Hz)		115 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 18.0
Max. fuse indoor/outdoor (A)		15.0 / 20.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 310
Input (W)		325 / 826
Rated current (RLA/A)		2.40 / 5.65
Operation Range		
ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

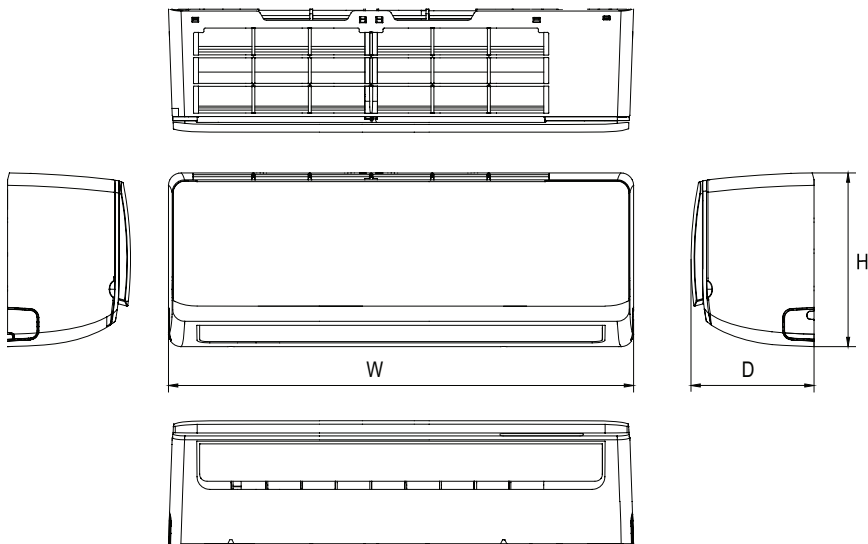
Piping

Refrigerant precharge (oz/kg)		33.51 / 0.95
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		82 / 25
Max. difference in height (ft/m)		49.2 / 15
System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.32
IDU	Fan motor model	ZKFP-20-8-113
	Fan motor qty (ea)	1
	Fan motor input (W)	50.0
	Fan speed (RPM)	1200 / 900 / 700
	Air flow, Turbo / H / M / L / Si (CFM)	400.2 / 323.7 / 282.5 / 235.4 / 211.9
	Approx. Air Throw Range ft (m)	23.0 / 7
	Noise level, Turbo / H / M / L / Si (dB(A))	44.0 / 36.5 / 32.5 / 21.0 / 17.5
	Net weight (lbs / kg)	23.15 / 10.5
	Gross weight (lbs / kg)	29.54 / 13.4
ODU	Fan motor model	ZKFN-34-10-1-9
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed, H / M / L (RPM)	850 / 700 / 600
	Air flow, Max (CFM)	1235.3
	Noise level, Max (dB(A))	52.5
	Net weight (lbs / kg)	64.59 / 29.3
	Gross weight (lbs / kg)	70.11 / 31.8
	Accessories	
BMS-WT2-XXD (8733962707)		Wired Wall Thermostat
BMS-DRY-XXD (8733969263)		24V Interface
BMS-PK2-7X1 (8733951011)		25ft Lineset
Standard Parts		
8-733-969-088		Remote Control

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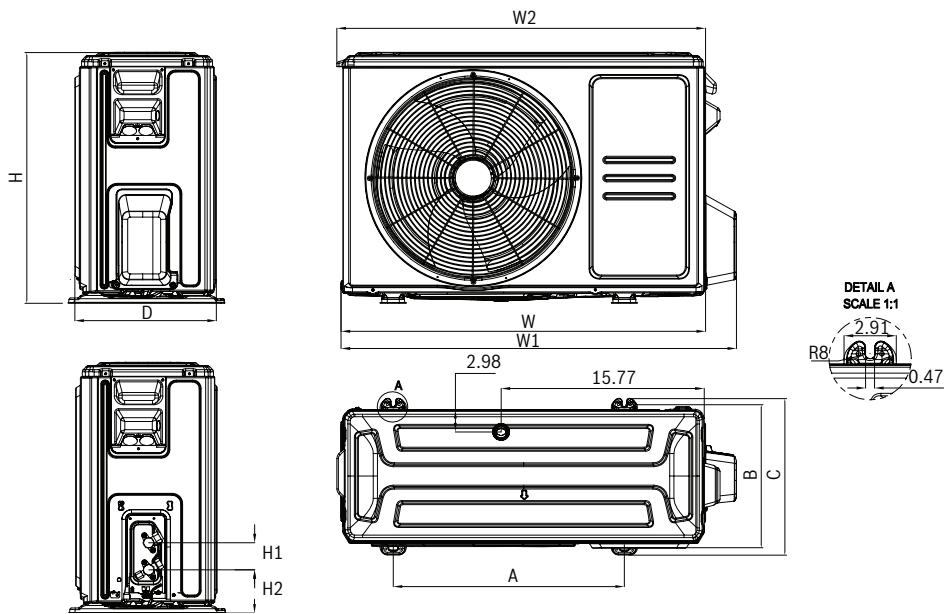


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAS012-0AHWXD	31.42	9.21	11.85

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS012-0CSXRD	30.12	11.93	21.85	32.9	30.87	2.36	3.67	17.8	11.26	12.36

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Cooling Performance - Indoor Dry Bulb Temperature at 400 CFM																					
ODU DB(°F)	IDU WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB(°F)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)
-13°F (-25.0°C)	TC	9700	9695	9834	9874	10146	10120	10112	10106	10105	10096	10423	10410	10400	10399	10389	10994	10991	10969	10965	10941
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.79	0.79	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.79	0.79	0.79	0.79	0.79
-4°F (-20.0°C)	TC	10446	10441	10590	10634	10927	10899	10890	10883	10882	10873	11225	11211	11200	11198	11188	11840	11837	11812	11808	11783
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81
0°F (-17.8°C)	TC	10793	10786	10986	11032	11335	11256	11243	11234	11233	11346	11586	11570	11556	11554	11548	12209	12207	12206	12207	12215
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.83	0.83	0.83	0.83	0.83
5°F (-15.0°C)	TC	11102	11140	11395	11448	11800	11623	11626	11626	11638	11812	11994	11998	12000	12000	12016	12698	12694	12701	12702	12708
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.84	0.84	0.84	0.84	0.84	0.85	0.85	0.86	0.86	0.86
14°F (-10.0°C)	TC	11303	11298	11528	11582	11941	11864	11854	11846	11845	11955	12271	12256	12243	12241	12229	13048	13044	13019	13014	12987
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81	0.81
17°F (-8.3°C)	TC	11344	11381	11644	11700	12080	11908	11913	11915	11925	12093	12315	12319	12324	12325	12340	13097	13093	13091	13094	13105
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.80	0.80	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80
23°F (-5.0°C)	TC	11322	11370	11665	11722	12109	11895	11899	11900	11916	12122	12305	12310	12313	12313	12333	13084	13080	13090	13091	13099
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.73	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.75	0.75	0.75	0.75	0.75	0.76	0.76	0.76	0.76	0.76
35°F (1.7°C)	TC	11385	11385	11452	11484	11855	11917	11917	11918	11918	11950	12290	12289	12290	12290	12292	12982	12979	12979	12978	12979
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.58	0.58	0.59	0.59	0.61	0.62	0.62	0.62	0.62	0.62	0.64	0.64	0.64	0.64	0.64	0.70	0.70	0.70	0.70	0.70
41°F (5.0°C)	TC	11071	11071	11159	11237	11569	11586	11586	11586	11586	11627	11947	11946	11947	11947	11948	12618	12614	12614	12613	12615
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.54	0.54	0.54	0.55	0.57	0.57	0.57	0.57	0.57	0.57	0.60	0.60	0.60	0.60	0.60	0.65	0.65	0.65	0.65	0.65
47°F (8.3°C)	TC	10756	10763	10908	10956	11281	11255	11255	11255	11255	11307	11604	11603	11604	11604	11604	12254	12251	12250	12249	12251
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.49	0.49	0.50	0.50	0.53	0.52	0.52	0.52	0.52	0.53	0.55	0.55	0.55	0.55	0.55	0.60	0.60	0.60	0.60	0.60
50°F (10.0°C)	TC	10650	10943	11312	11378	11819	11251	11257	11335	11371	11830	11690	11705	11711	11712	11825	12496	12529	12561	12565	12580
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
59°F (15.0°C)	TC	10907	11290	11690	11761	12240	11540	11556	11687	11774	12252	12016	12028	12032	12050	12262	12903	12929	12954	12957	12968
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.56	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
68°F (20.0°C)	TC	11157	11633	12065	12141	12656	11811	11838	12078	12154	12669	12322	12331	12353	12364	12679	13277	13302	13323	13325	13332
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.57	0.58	0.58	0.58	0.59	0.58	0.58	0.58	0.58	0.59	0.58	0.58	0.58	0.58	0.59	0.59	0.59	0.59	0.59	0.59
77°F (25.0°C)	TC	11396	11944	12412	12495	13053	12081	12119	12425	12509	13066	12628	12642	12675	12689	13077	13639	13680	13712	13715	13728
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.59	0.59	0.60	0.60	0.60	0.59	0.59	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.61	0.60	0.60	0.60	0.60
82°F (27.8°C)	TC	11463	12049	12535	12620	13197	12148	12195	12546	12634	13211	12713	12729	12768	12784	13223	13745	13797	13833	13836	13852
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.66	0.66	0.66	0.66	0.67	0.66	0.66	0.66	0.66	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67
86°F (30.0°C)	TC	11476	12049	12541	12629	13217	12102	12158	12555	12643	13231	12676	12690	12737	12755	13242	13723	13776	13808	13812	13825
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.78	0.79	0.79	0.79	0.80	0.79	0.79	0.79	0.79	0.80	0.79	0.79	0.79	0.79	0.80	0.80	0.80	0.80	0.80	0.80
95°F (35.0°C)	TC	11417	11479	11850	11920	12391	12134	12141	12145	12159	12407	12655	12658	12665	12667	12688	13636	13631	13634	13638	13656
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	1.03	1.03	1.04	1.05	1.07	1.06	1.06	1.06	1.06	1.07	1.08	1.08	1.08	1.08	1.08	1.13	1.13	1.13	1.13	1.13
104°F (40.0°C)	TC	10607	10704	11107	11174	11620	11276	11282	11302	11314	11634	11757									

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Heating Performance - Indoor Dry Bulb Temperature at 400 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	3881	3803	3784	3769	3735	0.75	0.76	0.76	0.76	0.77
-4°F(-20.0°C)	4486	4396	4374	4357	4317	0.82	0.83	0.83	0.83	0.83
0°F(-17.8°C)	5949	5830	5801	5778	5726	0.84	0.85	0.85	0.85	0.86
5°F(-15.0°C)	7779	7623	7585	7555	7487	0.87	0.88	0.88	0.88	0.89
14°F(-10.0°C)	8532	8361	8319	8286	8211	0.99	1.00	1.00	1.01	1.01
17°F(-8.3°C)	8783	8607	8564	8530	8453	1.03	1.04	1.04	1.05	1.05
22°F(-5.6°C)	9440	9251	9205	9168	9085	1.05	1.05	1.06	1.06	1.06
27°F(-2.8°C)	10097	9895	9846	9806	9718	1.06	1.07	1.07	1.07	1.08
32°F(0.0°C)	10754	10539	10486	10444	10350	1.07	1.08	1.08	1.09	1.09
35°F(1.7°C)	11148	10925	10871	10827	10730	1.08	1.09	1.09	1.09	1.10
37°F(2.8°C)	11640	11407	11350	11305	11203	1.08	1.09	1.09	1.09	1.10
42°F(5.6°C)	12869	12612	12548	12498	12386	1.08	1.09	1.09	1.09	1.10
44.6°F(7.0°C)	13361	13093	13028	12976	12859	1.08	1.09	1.09	1.09	1.09
47°F(8.3°C)	14098	13816	13747	13692	13569	1.08	1.08	1.09	1.09	1.09
52°F(11.1°C)	14161	13878	13809	13753	13630	1.01	1.01	1.02	1.02	1.02
57°F(13.9°C)	14225	13940	13870	13815	13691	0.93	0.94	0.94	0.95	0.95
59°F(15.0°C)	14250	13965	13895	13840	13715	0.91	0.91	0.91	0.92	0.92
62°F(16.7°C)	14288	14002	13932	13877	13752	0.86	0.87	0.87	0.87	0.88
64.4°F(18.0°C)	14318	14032	13962	13906	13781	0.83	0.84	0.84	0.84	0.84
69.8°F(21.0°C)	14387	14099	14029	13973	13847	0.75	0.76	0.76	0.76	0.76
75.2°F(24.0°C)	14455	14166	14095	14039	13913	0.68	0.68	0.68	0.68	0.69

Bolded values were tested. All other values are simulated and for reference only.